

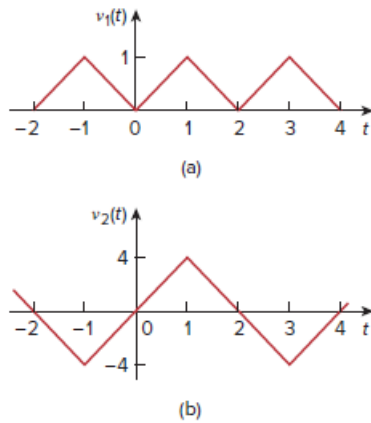
Problem

The waveform in Fig. 17.55(a) has the following Fourier series:

$$v_1(t) = \frac{1}{2} - \frac{4}{\pi^2} \left(\cos \pi t + \frac{1}{9} \cos 3\pi t + \frac{1}{25} \cos 5\pi t + \dots \right) \text{ V}$$

Obtain the Fourier series of $v_2(t)$ in Fig. 17.55(b).

Figure 17.55:



Step-by-step solution

Step 1 of 3

Refer to Figure 17.55 in the textbook.

The Fourier series of Figure 17.55(a) is,

$$v_1(t) = \frac{1}{2} - \frac{4}{\pi^2} \left(\cos \pi t + \frac{1}{9} \cos 3\pi t + \frac{1}{25} \cos 5\pi t + \dots \right) \text{ V}$$

Shift $v_2(t)$ by 1 along the vertical axis.

That is,

$$v_2^*(t) = v_2(t) + 1 \quad \dots (1)$$

Draw the waveform for $v_2^*(t)$.

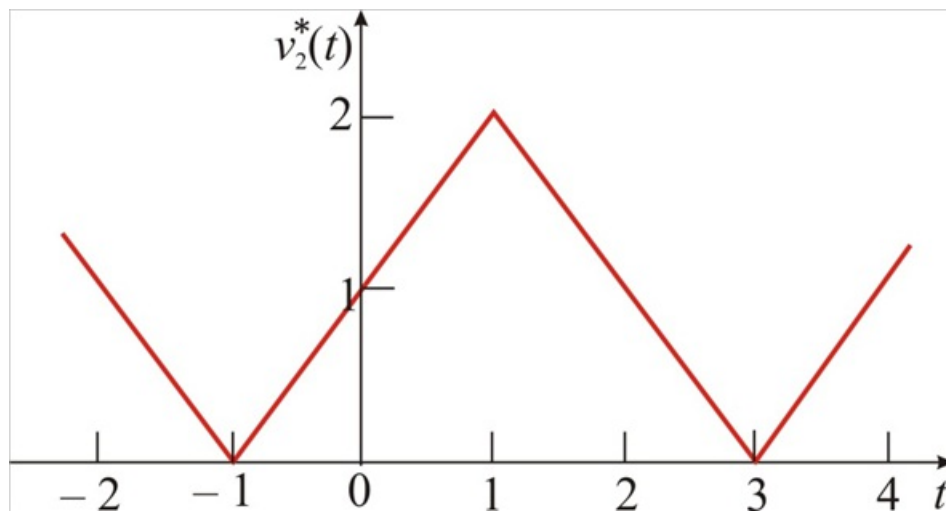


Figure 1

Step 2 of 3

Write the relation between $v_2^*(t)$ and $v_1(t)$.

$$v_2^*(t) = 2v_1\left(\frac{t+1}{2}\right)$$

Substitute $v_2(t)+1$ for $v_2^*(t)$.

$$2v_1\left(\frac{t+1}{2}\right) = v_2(t) + 1$$

$$v_2(t) = -1 + 2v_1\left(\frac{t+1}{2}\right) \dots\dots (2)$$

Step 3 of 3

Substitute $\frac{t+1}{2}$ for t in the expression $v_1(t)$.

$$\begin{aligned} v_1\left(\frac{t+1}{2}\right) &= \frac{1}{2} - \frac{4}{\pi^2} \left(\cos \pi \left(\frac{t+1}{2} \right) + \frac{1}{9} \cos 3\pi \left(\frac{t+1}{2} \right) + \frac{1}{25} \cos 5\pi \left(\frac{t+1}{2} \right) + \dots\dots \right) \\ &= \frac{1}{2} - \frac{4}{\pi^2} \left(\cos \left(\frac{\pi t}{2} + \frac{\pi}{2} \right) + \frac{1}{9} \cos \left(\frac{3\pi t}{2} + \frac{3\pi}{2} \right) + \frac{1}{25} \cos \left(\frac{5\pi t}{2} + \frac{5\pi}{2} \right) + \dots\dots \right) \\ &= \frac{1}{2} - \frac{4}{\pi^2} \left(-\sin \left(\frac{\pi t}{2} \right) + \frac{1}{9} \sin \left(\frac{3\pi t}{2} \right) - \frac{1}{25} \sin \left(\frac{5\pi t}{2} \right) + \dots\dots \right) \end{aligned}$$

Substitute $\frac{1}{2} - \frac{4}{\pi^2} \left(-\sin \left(\frac{\pi t}{2} \right) + \frac{1}{9} \sin \left(\frac{3\pi t}{2} \right) - \frac{1}{25} \sin \left(\frac{5\pi t}{2} \right) + \dots\dots \right)$ for $v_1\left(\frac{t+1}{2}\right)$ in equation (2).

$$\begin{aligned} v_2(t) &= -1 + 2 \left[\frac{1}{2} - \frac{4}{\pi^2} \left(-\sin \left(\frac{\pi t}{2} \right) + \frac{1}{9} \sin \left(\frac{3\pi t}{2} \right) - \frac{1}{25} \sin \left(\frac{5\pi t}{2} \right) + \dots\dots \right) \right] \\ &= -1 + 1 - \frac{8}{\pi^2} \left(-\sin \left(\frac{\pi t}{2} \right) + \frac{1}{9} \sin \left(\frac{3\pi t}{2} \right) - \frac{1}{25} \sin \left(\frac{5\pi t}{2} \right) + \dots\dots \right) \\ &= -\frac{8}{\pi^2} \left(-\sin \left(\frac{\pi t}{2} \right) + \frac{1}{9} \sin \left(\frac{3\pi t}{2} \right) - \frac{1}{25} \sin \left(\frac{5\pi t}{2} \right) + \dots\dots \right) \end{aligned}$$

Thus, the Fourier series of $v_2(t)$ is $\boxed{-\frac{8}{\pi^2} \left(-\sin \left(\frac{\pi t}{2} \right) + \frac{1}{9} \sin \left(\frac{3\pi t}{2} \right) - \frac{1}{25} \sin \left(\frac{5\pi t}{2} \right) + \dots\dots \right) \text{ V}}$